LABORATORY OBSERVATION / RECORD

Course Full Stack Web Development
Course Code 23CM4121
Experiment No. 6
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1. TITLE

Create a Functional Component and a Class Component using React

2. AIM

To create a functional component and a class component in React, and to display both of them inside the main App component of a React application.

3. OBJECTIVE

The objectives of this experiment are:

- To understand what a component is in React and why components are used.
- To create a functional component using a JavaScript function that returns JSX.
- To create a class component by extending React.Component and using the render() method.
- To export components using export default and import them into other files.
- To use the components inside App.jsx as JSX tags.
- To run the application and verify the output in the browser.

4. THEORY

React is a JavaScript library used to build user interfaces. A React application is built by combining small, reusable pieces called components. Each component returns JSX, which describes what should appear on the screen.

Functional component: It is a plain JavaScript function that returns JSX. It is the simplest way to write a component and is the style mostly used in modern React. In this experiment, FunctionalComponent returns a div with an h2 heading and a paragraph.

Class component: It is a JavaScript class that extends React.Component and must contain a render() method, which returns the JSX to be displayed. In this experiment, ClassComponent returns a div with an h2 heading and a paragraph.

JSX: JSX is a syntax extension that lets us write HTML-like tags inside JavaScript. A component name must start with a capital letter, for example <FunctionalComponent />, so that React treats it as a component.

Export and import: export default makes a component available to other files, and import brings it into App.jsx so that it can be used as a tag.

5. ALGORITHM / PROCEDURE

- Create a new React project (for example, using Vite) and open it in a code editor.
- Create the file Functionalcomponent.jsx and define the function FunctionalComponent that returns JSX.
- Export FunctionalComponent using export default.
- Create the file Classcomponent.jsx, import React, and define the class ClassComponent that extends React.Component.
- Add a render() method in ClassComponent that returns JSX, and export it using export default.
- Open App.jsx and import App.css, FunctionalComponent and ClassComponent.
- In the App function, return a div that contains an h1 heading, <FunctionalComponent /> and <ClassComponent />.
- Export App using export default.
- Run the project using npm run dev and open the local URL shown in the terminal.
- Verify that the heading and both components are displayed in the browser.

6. PROGRAM

Functionalcomponent.jsx

```
function FunctionalComponent() {
  return (
    <div>
      <h2>Functional Component</h2>
      <p>Hello! This is a functional component.</p>
    </div>
  );
}

export default FunctionalComponent;
```

Classcomponent.jsx

```
import React from 'react';

class ClassComponent extends React.Component {
  render() {
    return (
      <div>
        <h2>Class Component</h2>
        <p>Hello! This is a class component.</p>
      </div>
    );
  }
}

export default ClassComponent;
```

App.jsx

```
import './App.css'

import FunctionalComponent from './Functionalcomponent';
import ClassComponent from './Classcomponent';
```

```
function App() {
  return (
    <div>
      <h1>React Components Demo</h1>

      <FunctionalComponent />

      <ClassComponent />
    </div>
  );
}

export default App;
```

7. CODE EXPLANATION

Code	Explanation
import React from 'react'	Imports the React library so that the class can extend React.Component.
function FunctionalComponent()	Defines a functional component as a JavaScript function.
class ClassComponent extends React.Component	Defines a class component by extending the React.Component base class.
render()	Method of a class component that returns the JSX to be displayed.
return (...)	Returns the JSX (HTML-like markup) that the component displays.
, <h2>, <p>	JSX tags that create the container, the heading and the paragraph.
export default	Makes the component available for import in other files.
import './App.css'	Loads the stylesheet used by the App component.
import FunctionalComponent from './Functionalcomponent'	Imports the functional component into App.jsx.
import ClassComponent from './Classcomponent'	Imports the class component into App.jsx.
function App()	Main (root) component that combines the heading and both components.
<FunctionalComponent />	Uses the functional component as a JSX tag; its output appears here.
<ClassComponent />	Uses the class component as a JSX tag; its output appears here.

8. TEST CASES AND EXECUTION DATA

Each part of the application was executed and the actual output was recorded against the expected output.

Test Case	Component	Action	Expected Result	Actual Result	Status
TC-01	Project	Run npm run dev	Development server starts	Local: http://localhost:5173/ shown in terminal	Pass

Test Case	Component	Action	Expected Result	Actual Result	Status
TC-02	App.jsx	Open http://localhost:5173/	Page displays the heading	React Components Demo heading displayed	Pass
TC-03	FunctionalComponent	Check the first component	h2 and paragraph of the functional component shown	Functional Component / Hello! This is a functional component.	Pass
TC-04	ClassComponent	Check the second component	h2 and paragraph of the class component shown	Class Component / Hello! This is a class component.	Pass
TC-05	App.jsx	Check the order of output	Functional component first, then class component	Both components displayed in the same order as in App.jsx	Pass

9. OUTPUT

The application was run using the Vite development server and the output was checked in the browser. The output for each test case is given below.

Output 1 (TC-01): Server start

```
$ npm run dev
  VITE ready
  Local:   http://localhost:5173/
```

Output 2 (TC-02 to TC-05): Browser output

The browser shows the page with the heading, the functional component and the class component as given below. (Paste your browser screenshot here.)

```
React Components Demo

Functional Component
Hello! This is a functional component.

Class Component
Hello! This is a class component.
```

10. RESULT

Thus, a functional component and a class component were successfully created using React. Both components were exported, imported into App.jsx and displayed in the browser under the heading "React Components Demo". All test cases were executed and passed.